

# Ceremonial Entheogen Use in Puerto Rico: A Descriptive Survey of Participants and Facilitators

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## Abstract

Ceremonial use of entheogens occurs in Puerto Rico outside clinical and regulatory structures, and the practices surrounding it have not been described. We conducted an anonymous, bilingual, cross-sectional online survey of adults aged 21 and over reporting involvement in entheogen-assisted spiritual ceremonies in Puerto Rico, recruited by convenience and chain referral between February and June 2026. Of 100 submissions, 73 respondents identified a role:

7 72 as ceremony participants, 8 as facilitators, and 7 as both. Respondents were middle-aged  
8 (mean 42.4 years), predominantly university-educated, and almost all resident in Puerto Rico.  
9 Most recent ceremonies were typically single, multi-hour events in natural settings involving  
10 psilocybin mushrooms or ayahuasca, and satisfaction was strongly ceiling-weighted (median  
11 6 of 6). Preparation was near-universal among respondents answering, while pre-ceremony  
12 screening was not, and the small facilitator subsample described training and emergency plan-  
13 ning as informal rather than credentialed. Five of 67 participants answering reported non-  
14 consensual physical contact during a ceremony. A pre-declared comparison of any preparation  
15 against none on satisfaction was directionally positive but rests on five unprepared respondents  
16 and is exploratory. These findings describe this sample only and support no prevalence or  
17 causal claim.

18 **Keywords:** entheogens; psychedelics; ceremonial use; Puerto Rico; harm reduction; survey re-  
19 search

## 20 1 Introduction

21 Ceremonial use of entheogens—psilocybin-containing mushrooms, ayahuasca, and related prepa-  
22 rations taken in structured group or guided settings—has become substantially more visible across  
23 the Americas over the past two decades, largely outside the clinical-trial infrastructure that gen-  
24 erates most of the contemporary research literature (Golden et al., 2022; Carvalho et al., 2025).  
25 The resulting evidence base is asymmetric. Randomized clinical research characterizes carefully  
26 screened participants receiving standardized doses under monitored conditions, and its samples  
27 remain narrow (Hughes and Garcia-Romeu, 2024; Marcus and Rush, 2026), while community  
28 ceremonial practice is documented mainly through ethnographic and qualitative work, with com-  
29 paratively few survey-level descriptions of who takes part and under what conditions (Carvalho  
30 et al., 2025).

31 Puerto Rico occupies an unusual position in this landscape. It is a Spanish-speaking Caribbean  
32 jurisdiction in which psilocybin, DMT, and related compounds are Schedule I substances under the

federal Controlled Substances Act (United States Congress, 1970) and under the territory’s own aligned statute (Legislative Assembly of Puerto Rico, 1971), and which sits outside the state-level decriminalization and regulated-access reforms that have reshaped the legal environment in parts of the United States (Siegel et al., 2023). It is also a setting with a long-documented tradition of spiritually framed healing practice—espiritismo and related practices—that has been described in the clinical and public health literature for more than four decades (Koss, 1980; Comas-Díaz, 1981; Bird and Canino, 1981; Zerrate et al., 2022).

Local survey evidence on entheogen use is nonetheless minimal. One published Puerto Rico-specific survey exists, conducted by members of the present research team, but it is narrower in scope: it characterizes psilocybin use patterns, motivations, and personality correlates rather than ceremonial practice, and addresses neither facilitation nor safety (Vélez Rodríguez et al., 2026). An unpublished thesis has examined university-community perceptions of psilocybin mushrooms (Doelter Morales and Martínez Miranda, 2020). Neither describes how ceremonies are actually conducted. Basic descriptive questions—who takes part, which substances are used, where and for how long ceremonies occur, whether participants are prepared or screened beforehand, and what adverse experiences are reported—therefore have no published local answer.

A descriptive characterization contributes in three ways. First, it establishes whether practices reported in a community sample resemble the harm-reduction structures assumed by clinical and retreat-based frameworks: preparation, pre-ceremony screening, and a plan for difficult experiences (Gorman et al., 2021; Pilecki et al., 2021; Dutton et al., 2025). Second, it identifies which safety outcomes—including boundary violations, shown elsewhere to be measurable in naturalistic samples (Kruger et al., 2025; Peluso et al., 2020)—occur at frequencies justifying measurement in a larger study. Third, it produces the instrument-level evidence needed before any confirmatory study is designed in this population.

Our objectives were to describe the sample and the ceremonial context, and to examine bivariate associations between preparation and reported ceremony outcomes and between participant-observable facilitator practices and perceived safety. A safety and adverse-event domain, absent

from the original protocol, was added during analysis. Every objective is descriptive. The design supports no statement about how common these practices are in Puerto Rico, and no statement about what any practice produces.

## **2 Methods**

### **2.1 Design, ethics, and consent**

This was a single-timepoint, cross-sectional, self-administered online survey of adults reporting involvement in entheogen-assisted spiritual ceremonies in Puerto Rico. The sample was non-probability and the analysis unweighted. The study was approved by the University of Puerto Rico Institutional Review Board (protocol 2509472552, approved 4 November 2025) and was fielded on KoboToolbox between 24 February and 9 June 2026. Participation was fully anonymous: no names, dates of birth, or other personal identifiers were collected. Respondents read an information sheet covering purpose, procedures, risks, voluntariness, and data retention, and indicated consent by clicking through to the questionnaire; no signed consent document was collected, consistent with the anonymous design. “Prefer not to answer” was offered on every item, withdrawal was permitted at any point, no compensation was offered, and mental-health support resources were listed. An optional future-contact form was linked separately, with no join key to questionnaire responses. Reporting follows STROBE for cross-sectional studies (Supplementary Material).

### **2.2 Recruitment and sample construction**

Recruitment used convenience sampling with chain referral, promoted online through Facebook, Instagram, Twitter/X, and WhatsApp, together with bulletin-board announcements at the University of Puerto Rico Río Piedras and Medical Sciences campuses. Inclusion criteria were age 21 or older, ability to read and write in Spanish, access to an internet-connected device, and prior participation in an entheogen-assisted spiritual ceremony in Puerto Rico. Persons under 21 or unable to

consent were excluded. The recruitment target of 100 responses was met.

Of the 100 submissions collected, 13 were abandoned before any substantive item and were excluded, leaving 87 respondents who reached the role-identification questions. Of those, 14 (16%) answered “no” to both role items, indicating no ceremony involvement, and are reported here as a screening yield rather than as a described subgroup. The analytic sample is the remaining  $N = 73$  respondents who identified a role. Roles are not a partition: 72 respondents identified as ceremony participants and 8 as facilitators, with 7 identifying as both and appearing in both role-specific analyses. Critically, the two branches are *unlinked*. The instrument carries no ceremony or facilitator identifier, so no participant can be connected to the facilitator who ran their ceremony, and facilitator-reported and participant-reported measures cannot be crossed. Figure 1 shows the full construction.

## 2.3 Instrument

No validated Puerto Rico–specific instrument for characterizing ceremonial entheogen use exists, so the questionnaire was developed by the research team, reviewed iteratively, professionally translated by direct translation, and given a final team review for cross-language consistency. It was deployed as a single bilingual XLSForm with Spanish as the default language and English available through the platform’s language switcher, rather than as two separate instruments; open-text responses were accordingly mixed-language.

The primary outcome, satisfaction with the most recent ceremony, asked “From 0 to 6, how pleasant was the ceremony?” with anchors from *very unpleasant* (0) to *very pleasant* (6), and was treated as ordinal. The primary exposure asked, “Aside from the facilitator’s instructions or recommendations, do you personally engage in preparation for these ceremonies?” (yes / sometimes / no), and therefore captures self-directed preparation only, not facilitator-provided instruction. Secondary participant-side measures covered in-ceremony comfort and safety (0–3), trust in the facilitator (0–4), prior knowledge of the facilitator’s background, whether the facilitator asked screening questions beforehand, and whether the respondent had spoken to a health professional

beforehand. The non-consensual contact item asked whether the respondent had ever experienced physical touch during a ceremony that felt unexpected or that they had not clearly agreed to beforehand, explicitly including moments of uncertainty or absent recall; its breadth exceeds a narrower assault framing and the resulting figure should be read accordingly. Facilitator-branch items covered training, screening approach, distress-response protocol, emergency planning, and crisis signs witnessed. Three substance variables were kept distinct and never merged: facilitator-led substances, participant substances across all ceremonies attended, and the substance used in the most recent ceremony. Attention checks were branch-specific. Verbatim wording and anchors for all key items appear in the Supplementary Material.

## **2.4 Missing data, structural skips, and suppression**

Denominators shift across items for two distinct reasons, kept separate throughout: structural skips arise from branch logic and are reported as their own row, while item nonresponse is the residual. Every figure carries its own base and no denominator is carried across sections. “Prefer not to answer” and “I’m not sure” were retained as substantive categories in all frequency tables, excluded from ordinal numeric tests because they carry no rank position, and treated as item-missing rather than scored as zero in composite indices. Subgroup cells containing fewer than five respondents are suppressed and are not back-calculated; counts of exactly five are reported exactly, a disclosure decision the research team made deliberately on the grounds that no respondent is identifiable from these aggregates.

Because the platform populated no duplicate-detection metadata and collected no IP or device information, duplicate submissions were screened manually on the respondent-chosen nickname and password. One pair shared both fields, and its earlier member was already excluded as an abandoned submission, so no analytic row required exclusion on duplicate grounds; repeated nicknames or passwords without a matching second field involved common Spanish words and were judged coincidental (Supplementary Material). Two respondents failed at least one attention check and were flagged rather than dropped, per protocol.

## 2.5 Statistical approach

The outcome distribution was strongly ceiling-weighted and several exposures had thin or unbalanced categories, so all tests were nonparametric: Mann–Whitney  $U$  for two-group ordinal comparisons, Kruskal–Wallis  $H$  for three or more groups, permutation  $\chi^2$  with 10,000 label permutations (seed 20260715) for thin contingency tables, and Fisher’s exact test for  $2 \times 2$  tables at very small  $n$ . Effect sizes were rank-biserial  $r_{rb}$  with the common-language effect size, epsilon-squared  $\varepsilon^2$ , and Cramér’s  $V$  respectively, with Wilson 95% confidence intervals for prevalences. The bias-corrected  $\varepsilon^2$  formula can return small negative values when an effect is indistinguishable from zero; these are displayed truncated to 0.00 (marked †), applied uniformly.

The preparation-to-satisfaction comparison was pre-declared and is reported uncorrected, with a tie-aware permutation test as a robustness check because the outcome is heavily tied. All other bivariate tests were assigned to four prespecified secondary and exploratory families—bivariate, substance-stratified, safety-domain, and legal-knowledge—corrected jointly within family by the Benjamini–Hochberg false-discovery-rate procedure and kept separate because they share neither exposures nor outcome domains. One multivariable proportional-odds model was fitted as an exploratory supplement. Analyses were conducted in Python; the full pipeline is available (see Data availability).

## 2.6 Protocol deviations

Three deviations are reported transparently. First, the original protocol specified a four-arm preparation  $\times$  integration design. This is not estimable from the collected data: the instrument contains no clean participant integration-received item, and the preparation split is badly unbalanced. Preparation was therefore examined against the outcome directly, and no integration measure is reconstructed anywhere in this paper.

Second, the protocol specified crossing facilitator training and protocol items against participant-perceived safety. Because those items are facilitator self-report from a different and unlinked set of respondents, crossing them would have required pairing unrelated people. Two

participant-answerable proxies were substituted—prior knowledge of the facilitator’s background and whether screening questions were asked—and facilitator self-report is presented separately and descriptively.

Third, the protocol specified a composite harm-reduction score mixing facilitator-side and participant-side items, which for the same reason could not validly be built; two branch-specific composites were constructed instead. Two skip-logic errors in the instrument were also identified and are documented in the Supplementary Material.

## 3 Results

### 3.1 Respondents

The analytic sample of 73 adults skewed middle-aged and highly educated (Table 1). Among the 72 respondents with a valid age, the mean was 42.4 years and the median 41 (SD 11.9, range 21–71, IQR 34–50). Of the 71 answering the gender item, 35 (49%) identified as women and 30 (42%) as men, with non-binary and prefer-not-to-answer cells suppressed. Of the 72 answering the education item, 34 (47%) reported a bachelor’s degree and 19 (26%) a master’s, and of the 72 answering the economic item, 44 (61%) described their situation as comfortable against 6 (8%) as difficult. Sixty-eight respondents (94% of the 72 answering) reported residing in Puerto Rico; residency was treated as a descriptive covariate rather than an inclusion filter, and the presence of respondents living elsewhere tempers any reading of the sample as describing Puerto Rico specifically.

Free-text engagement was substantial relative to sample size: 37 open-text fields yielded 285 non-empty answers, roughly three-quarters of them in Spanish, most often on an experience highlight (58 answers), opinions on safety (54), and preparation guidance received (45). A coding scaffold for the research team’s thematic analysis accompanies these data (Supplementary Material); the corpus is not interpreted here.

## 3.2 The ceremonial context

Participants reported extensive but not recent ceremonial histories. Among the 67 answering, the median number of lifetime ceremonies attended was 6 (mean 15.1, SD 35.7, range 1–250, IQR 3–12), a long right tail that makes the median the appropriate summary, while ceremonies in the past 12 months were far fewer in the same 67 (median 1, mean 1.3, SD 1.7, range 0–8, IQR 0–2). Of the 66 answering the recency item, 30 (45%) reported their last ceremony more than 12 months earlier.

Across all ceremonies attended, the substances most often reported by the 67 participants answering were magic mushrooms (50, 75%), ayahuasca (43, 64%), tobacco (38, 57%), and cacao (23, 34%), with DMT, LSD, MDMA, and ketamine less frequent (Figure 2). For the most recent ceremony specifically, individually reportable groups were magic mushrooms ( $n = 31$ ) and ayahuasca ( $n = 23$ ), with every remaining substance individually below the suppression floor and pooled into a combined group ( $n = 11$ ) under a prespecified rule.

The most recent ceremony was typically a discrete, multi-hour, outdoor event (Table 2). Of the 65 answering the format item, 48 (74%) reported a single ceremony rather than a retreat or series; of the 67 answering the place item, 37 (55%) reported a natural setting such as a forest, beach, or mountain; and of the 65 answering the duration item, 19 (29%) reported 4–6 hours and a further 19 (29%) reported an overnight ceremony of roughly 12 hours or more.

## 3.3 Preparation, screening, and prior knowledge

Preparation was near-universal among those answering (Table 3): of the 67 answering, 55 (82%) reported preparing and 6 (9%) preparing sometimes, against 6 (9%) reporting no self-directed preparation. This imbalance is why the four-arm design was abandoned and it constrains every preparation analysis below. Participant-observable facilitator practices were common but not uniform. Of the 67 answering, 54 (81%) reported knowing about the facilitator’s background or training beforehand and 8 (12%) knowing a little, with the remaining cells suppressed. On screening, of the 66 answering, 39 (59%) reported that the facilitator asked screening questions beforehand, 15

(23%) that none were asked, and 7 (11%) were unsure. Disclosure outside the ceremonial context was uncommon: of the 65 answering, 8 (12%) reported having spoken to a doctor, therapist, or other health professional about their participation beforehand, against 54 (83%) who had not.

A participant-side harm-reduction practice index counting these three practices (preparation, prior knowledge of the facilitator’s background, screening received; range 0–3) was constructed for the 54 participants with complete data on all three. Thirty-five (65%) reported all three and 16 (30%) reported two; *none* reported zero. Internal consistency was low ( $\alpha = 0.14$ , 95% CI  $-0.36$ – $0.47$ ), indicating that the items capture distinct protective practices rather than one latent construct, so the score is read as a count of practices present rather than as a validated scale.

### 3.4 Experience and perceived safety

Reported satisfaction was overwhelmingly positive. Of the 72 participants, 63 answered the satisfaction item, and the distribution was strongly ceiling-weighted with a median of 6 and the mode at the scale maximum (Figure 3). Perceived in-ceremony comfort and safety was similarly high and similarly undifferentiated, returning a median of 3 in every subgroup tested and differing neither by prior knowledge of the facilitator’s background ( $H = 2.24$ ,  $p = 0.326$ ,  $\varepsilon^2 = 0.00$ ) nor by whether screening questions were asked ( $H = 2.40$ ,  $p = 0.301$ ,  $\varepsilon^2 = 0.01$ ). Trust in the facilitator was the one participant-side comparison differing at the uncorrected level: median 4 (IQR 3–4) among the 54 who knew the facilitator’s background beforehand against 3 (IQR 2–3) among the 8 who knew a little, with a suppressed third group ( $H = 8.15$ ,  $p = 0.017$ ,  $\varepsilon^2 = 0.10$ ); it did not survive correction ( $q = 0.102$ ). Satisfaction did not differ significantly by most-recent-ceremony substance (ayahuasca median 5, mushrooms median 6, combined group median 6;  $H = 4.98$ ,  $p = 0.083$ ,  $q = 0.166$ ).

The safety domain produced the sample’s most consequential descriptive finding. Of the 67 participants answering the non-consensual contact item, 61 (91%) reported no such contact and 5 (7%) reported that it occurred, with a small uncertain group suppressed; the within-sample proportion was 7.5% (Wilson 95% CI 3.2–16.3). This describes these 67 respondents and nothing

beyond them. Non-consensual contact was associated with prior knowledge of the facilitator's background at the uncorrected level (permutation  $\chi^2 = 18.90$ ,  $p = 0.047$ ,  $V = 0.38$ ,  $n = 67$ ) but not with screening (permutation  $\chi^2 = 13.64$ ,  $p = 0.094$ ,  $V = 0.32$ ,  $n = 66$ ); neither survived correction (both  $q = 0.236$ ), nor did the comparison by substance ( $p = 0.486$ ,  $q = 0.486$ ) or by the harm-reduction index ( $p = 0.641$ ,  $q = 0.641$ ). The research team elected to publish these cross-tabulations under the same suppression convention applied to every other thin-cell table rather than withhold them: withholding would remove the only evidence this study can offer on whether such reports cluster with any measured practice, and the masking convention leaves no cell small enough to plausibly identify a respondent. Every result in this domain is hypothesis-generating and is powered neither to confirm nor to rule out an association.

One measurement observation belongs here: an item on perceived positive life impact had zero variance, with all 67 respondents answering selecting the same option.

### 3.5 Facilitator practice

Eight respondents identified as facilitators. This is a different, unlinked set from the participants above, nearly every category cell falls at or below the suppression floor, and no group comparison at this base is meaningful; the material is descriptive context only. Preparation for the role was informal: 6 of the 8 (75%) reported personal experience and self-study, with mentorship and family or ancestral tradition each reported by suppressed numbers, and *no* facilitator reported formal certification or academic coursework. Six of 8 (75%) reported an initial conversation or interview with prospective participants, and 5 of 8 (62%) reported using singing, music, or sound to guide someone through a difficult experience. On emergency planning, all responses at a base of 7 fell into informal or conditional categories, with no facilitator selecting the formal option. Seven facilitators answered the crisis-signs item; individual sub-items, including suicidal ideation and panic attack, are each suppressed and establish only that such events were witnessed and fall within scope. Association tests within this branch returned undefined odds ratios at  $n = 7$  (Supplementary Material).

### 3.6 Preparation and reported satisfaction

Among the 63 participants answering the satisfaction item, 58 reported some preparation (median 6, IQR 5–6) and 5 reported none (median 4, IQR 3–5; Figure 4); one further respondent reporting no preparation did not answer the satisfaction item, which is why this base is 5 rather than the 6 in Table 3. The pre-declared Mann–Whitney comparison returned  $U = 218.0$ ,  $p = 0.041$ ,  $r_{rb} = 0.50$ , and a common-language effect size of 0.75, with preparation associated with higher reported satisfaction; a tie-aware permutation test gave  $p = 0.024$ . The structural weakness of this comparison, however, precedes its result: the no-preparation group contains five respondents. The prespecified three-level sensitivity analysis was non-significant ( $H = 4.24$ ,  $p = 0.120$ ,  $\varepsilon^2 = 0.04$ ;  $q = 0.359$ ), the practice index was unassociated with satisfaction ( $H = 2.09$ ,  $p = 0.352$ ;  $q = 0.422$ ), and an exploratory proportional-odds model adjusting preparation for age and facilitator trust placed preparation short of conventional significance (OR 5.21, 95% CI 0.84–32.41,  $p = 0.077$ ) while trust reached it (OR 2.20, 95% CI 1.06–4.53,  $p = 0.033$ ) at roughly 12 cases per estimated parameter (Supplementary Material). The comparison is suggestive and exploratory, not confirmatory. No test in any corrected family was significant at  $q < 0.05$ .

## 4 Discussion

### 4.1 A structurally informal practice landscape

The most stable picture this sample supports is a practice organized around informal, relationally transmitted expertise rather than credentialed or documented protocol. Among the eight facilitators who self-reported, training came from personal experience, self-study, mentorship, and family tradition; none reported formal certification or academic coursework, and emergency planning was uniformly informal or conditional. This sits at some distance from the structured preparation, screening, and difficult-experience protocols that harm-reduction frameworks for psychedelic settings assume (Gorman et al., 2021; Pilecki et al., 2021; Dutton et al., 2025), and it is consistent

with survey evidence that people who use psychedelics support clearer practice standards (Kruger et al., 2023).

On the participant side the informality reads differently. Preparation was near-universal among those answering, but roughly a quarter of participants answering the screening item reported that no screening questions were asked and a further group was uncertain. The asymmetry is worth separating: preparation is something participants largely do, whether self-directed or instructed, whereas screening must be done *to* them, and was substantially less consistent in what these respondents reported. If that asymmetry holds in a larger sample it would locate the intervention target on the facilitator side—but this sample cannot establish whether it holds, and eight facilitators cannot characterize screening practice.

The demographic profile—middle-aged, university-educated, economically comfortable—resembles that of ceremony-attender cohorts described elsewhere (Teixeira et al., 2026; Pagni et al., 2025), of large international psychedelic-survey samples (Kopra et al., 2023; Robinson et al., 2026), and of the psilocybin-specific Puerto Rico survey conducted by this team (Vélez Rodríguez et al., 2026). That convergence is best read as evidence about who responds to online psychedelic surveys, not about who attends ceremonies.

## 4.2 A ceiling that limits what satisfaction can reveal

Reported satisfaction clustered at the top of its scale, and a perceived-life-impact item had no variance at all. This is a substantive finding—these respondents overwhelmingly described their most recent ceremony positively, consistent with naturalistic cohorts reporting predominantly positive appraisals (Nayak et al., 2023; Ruffell et al., 2021)—but it also constrains the analysis severely. An outcome with almost no variance in the lower range cannot discriminate between groups, and the pattern of null results across the corrected families is consistent with that constraint rather than with the absence of any association. Ceiling behavior is a known property of self-report scales for hallucinogen experience rather than a peculiarity of this instrument (Bouso et al., 2016). The design implication is that satisfaction, measured this way, is close to exhausted as an outcome

in this population; future work needs measures with headroom in both directions before group comparisons on experience quality can be informative.

### **4.3 Safety signals the design can identify but not quantify**

Five of 67 participants answering reported non-consensual physical contact, and facilitators reported having witnessed crisis signs, though every individual sub-item was suppressed and no rate was estimable at a base of seven. These two findings come from different, unlinked reporters and cannot be combined. That such reports appear at all is consistent with the only survey-based evidence on boundary violations in naturalistic psychedelic settings (Kruger et al., 2025) and with the community harm-reduction literature that has developed in response (Peluso et al., 2020). The item used here was deliberately broad, covering unexpected touch and uncertain or unremembered consent, so the figure should not be read against a narrower legal or clinical standard.

The study establishes that these events occur among respondents and that an instrument of this kind can capture them. It cannot establish how often they occur, in whom, or under what conditions. The cross-tabulations against participant-side proxies are powered neither to confirm nor to rule out an association, and the one comparison reaching uncorrected significance did not survive correction; treating any of them as an actionable signal would overstate them considerably. The safety domain justified its inclusion by identifying a measurement priority, not by producing an estimate.

### **4.4 The preparation finding as a design lesson**

The preparation-to-satisfaction comparison is this study's pre-declared primary analysis and its most fragile result, and those two facts should be held together rather than traded off. The direction was positive and the effect sizes were not small, but the no-preparation group contained five respondents answering the outcome, the three-level sensitivity analysis was non-significant, and the adjusted model placed preparation short of significance with a confidence interval spanning nearly two orders of magnitude. Nothing here establishes the finding.

The more durable contribution is diagnostic. The preparation split was unbalanced enough to dissolve the original four-arm design and it remains the binding constraint on the primary analysis. Any successor study intending to examine preparation will need either a sampling strategy that reaches unprepared participants in usable numbers or a question that does not depend on that contrast. Reporting the deviation rather than absorbing it quietly is what makes the constraint visible to the next design.

## 4.5 Limitations

The sample was non-probability, unweighted, and self-selected into an online survey; the concentration in higher education categories and favorable economic self-ratings describe who responded, not who participates. Nothing here supports a statement about how common ceremonial entheogen use is in Puerto Rico or about people who use entheogens in Puerto Rico generally. Roughly 6% of those answering the residency item lived outside Puerto Rico, further tempering any Puerto Rico-specific reading.

Role subsamples were small and unevenly sized, and the facilitator branch of eight supports no rate and no association test. Participant and facilitator respondents are unlinked, which is not a minor analytic inconvenience: it eliminated the protocol's integration-proxy comparison, its perceived-safety-by-training analysis, and its mixed harm-reduction composite, and it means no statement in this paper connects what facilitators reported doing to what participants reported experiencing.

All measures were single-timepoint self-report, and for most respondents the most recent ceremony was distant—30 of 66 answering reported more than 12 months—so recall error is plausible and unmeasurable here. The instrument contains no integration measure, is unvalidated, and produced mixed first reliability evidence alongside two documented skip-logic errors (Supplementary Material).

## 4.6 Conclusion

Within a self-selected sample of 73 adults, ceremonial entheogen use in Puerto Rico was described by a middle-aged, university-educated, overwhelmingly Puerto Rico–resident group whose most recent ceremonies were typically single, multi-hour, outdoor events involving psilocybin mushrooms or ayahuasca, and who described those ceremonies in strongly positive terms. Preparation was near-universal while pre-ceremony screening was not, and the small facilitator subsample described training and emergency planning as informal rather than credentialed. Five of 67 participants answering reported non-consensual physical contact, and facilitators reported witnessing crisis events at a base too small for any rate to be estimated. These findings describe this sample alone. They support no estimate of prevalence and no inference about what any practice produces, and their principal value lies in identifying which measures, sampling strategies, and instrument corrections a properly powered successor study would require.

## Author contributions

YOOO conceived the study, led protocol development and ethics submission, and supervised the project. YOOO, JCVR, JMHT, JMT, PRF, CR, ANGR, and AIRM contributed to instrument development and review. JCVR led data curation, formal analysis, and manuscript drafting. JMHT contributed to methodology and analysis review. JMT, PRF, and CR contributed to interpretation and clinical/community context. ANGR and AIRM contributed to recruitment and data collection. All authors reviewed and approved the final manuscript.

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## Competing interests

The authors declare no competing interests.

## Data availability

The analysis code that generates every result reported here is available from the corresponding author. De-identified survey data are available from the corresponding author on reasonable request, subject to the terms of the approved protocol.

## Generative AI use declaration

Large language model tooling (Anthropic Claude) was used to assist with statistical analysis code development and with drafting and editing of this manuscript. All analyses were specified by the authors, all outputs were verified against the source data by the authors, and the authors take full responsibility for the content of this article.

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## Tables and Figures

Table 1: Sociodemographic characteristics of the analytic sample ( $N = 73$ ). Bases differ by item; cells with  $n < 5$  are suppressed and are not back-calculated. Percentages are of respondents answering.

| Characteristic                    | Category                                                | $n$ | % answering |
|-----------------------------------|---------------------------------------------------------|-----|-------------|
| Age, years ( $n = 72$ )           | Mean 42.4 (SD 11.9); median 41 (IQR 34–50); range 21–71 |     |             |
| Gender ( $n = 71$ )               | Woman                                                   | 35  | 49          |
|                                   | Man                                                     | 30  | 42          |
|                                   | Non-binary                                              | < 5 | 6           |
|                                   | Prefer not to answer                                    | < 5 | 3           |
| Education ( $n = 72$ )            | Bachelor's degree                                       | 34  | 47          |
|                                   | Master's degree                                         | 19  | 26          |
|                                   | Associate's degree                                      | 7   | 10          |
|                                   | High school diploma / GED                               | 6   | 8           |
|                                   | Doctorate                                               | < 5 | 4           |
|                                   | Prefer not to answer                                    | < 5 | 4           |
| Economic situation ( $n = 72$ )   | Comfortable                                             | 44  | 61          |
|                                   | Tight                                                   | 20  | 28          |
|                                   | Difficult                                               | 6   | 8           |
|                                   | Prefer not to answer                                    | < 5 | 3           |
| Country of residence ( $n = 72$ ) | Puerto Rico                                             | 68  | 94          |
|                                   | United States                                           | < 5 | 4           |
|                                   | Spain                                                   | < 5 | 1           |

Table 2: Most-recent-ceremony context, participant-reported. Participant branch only (base 72); bases differ by item because structural skips are reported separately from item nonresponse. Cells with  $n < 5$  are suppressed.

| Item                                  | Category                                  | $n$ | % answering |
|---------------------------------------|-------------------------------------------|-----|-------------|
| Time since last ceremony ( $n = 66$ ) | More than 12 months ago                   | 30  | 45          |
|                                       | 2 to 6 months ago                         | 20  | 30          |
|                                       | 7 to 12 months ago                        | 11  | 17          |
|                                       | During the past month                     | < 5 | 6           |
| Place ( $n = 67$ )                    | Natural setting (forest, beach, mountain) | 37  | 55          |
|                                       | Private home                              | 15  | 22          |
|                                       | Retreat center or wellness facility       | 9   | 13          |
| Format ( $n = 65$ )                   | Single ceremony                           | 48  | 74          |
|                                       | Retreat or series of ceremonies           | 13  | 20          |
| Duration ( $n = 65$ )                 | 4 to 6 hours                              | 19  | 29          |
|                                       | Overnight (~12+ hours)                    | 19  | 29          |
|                                       | 7 to 12 hours                             | 14  | 22          |
|                                       | 1 to 3 hours                              | 7   | 11          |

Table 3: Participant-reported protective practices and safety outcomes. Bases are per item and differ because each is asked under its own skip condition. Cells with  $n < 5$  are suppressed; counts of exactly 5 are reported exactly.

| Item                                                   | Category                        | $n$ | % answering |
|--------------------------------------------------------|---------------------------------|-----|-------------|
| Self-directed preparation ( $n = 67$ )                 | Yes                             | 55  | 82          |
|                                                        | Sometimes                       | 6   | 9           |
|                                                        | No                              | 6   | 9           |
| Knew facilitator's background ( $n = 67$ )             | Yes                             | 54  | 81          |
|                                                        | A little                        | 8   | 12          |
|                                                        | No                              | < 5 | 6           |
|                                                        | Prefer not to answer            | < 5 | 1           |
| Facilitator asked screening questions ( $n = 66$ )     | Yes                             | 39  | 59          |
|                                                        | No                              | 15  | 23          |
|                                                        | Not sure                        | 7   | 11          |
|                                                        | Prefer not to answer            | 5   | 8           |
| Spoke to a health professional beforehand ( $n = 65$ ) | No                              | 54  | 83          |
|                                                        | Yes                             | 8   | 12          |
|                                                        | Prefer not to answer / not sure | < 5 | 5           |
| Non-consensual physical contact ( $n = 67$ )           | No                              | 61  | 91          |
|                                                        | Yes                             | 5   | 7           |
|                                                        | Not sure                        | < 5 | 1           |
| Harm-reduction practice index ( $n = 54$ )             | All three practices             | 35  | 65          |
|                                                        | Two practices                   | 16  | 30          |
|                                                        | One practice                    | < 5 | 6           |
|                                                        | No practices                    | 0   | 0           |

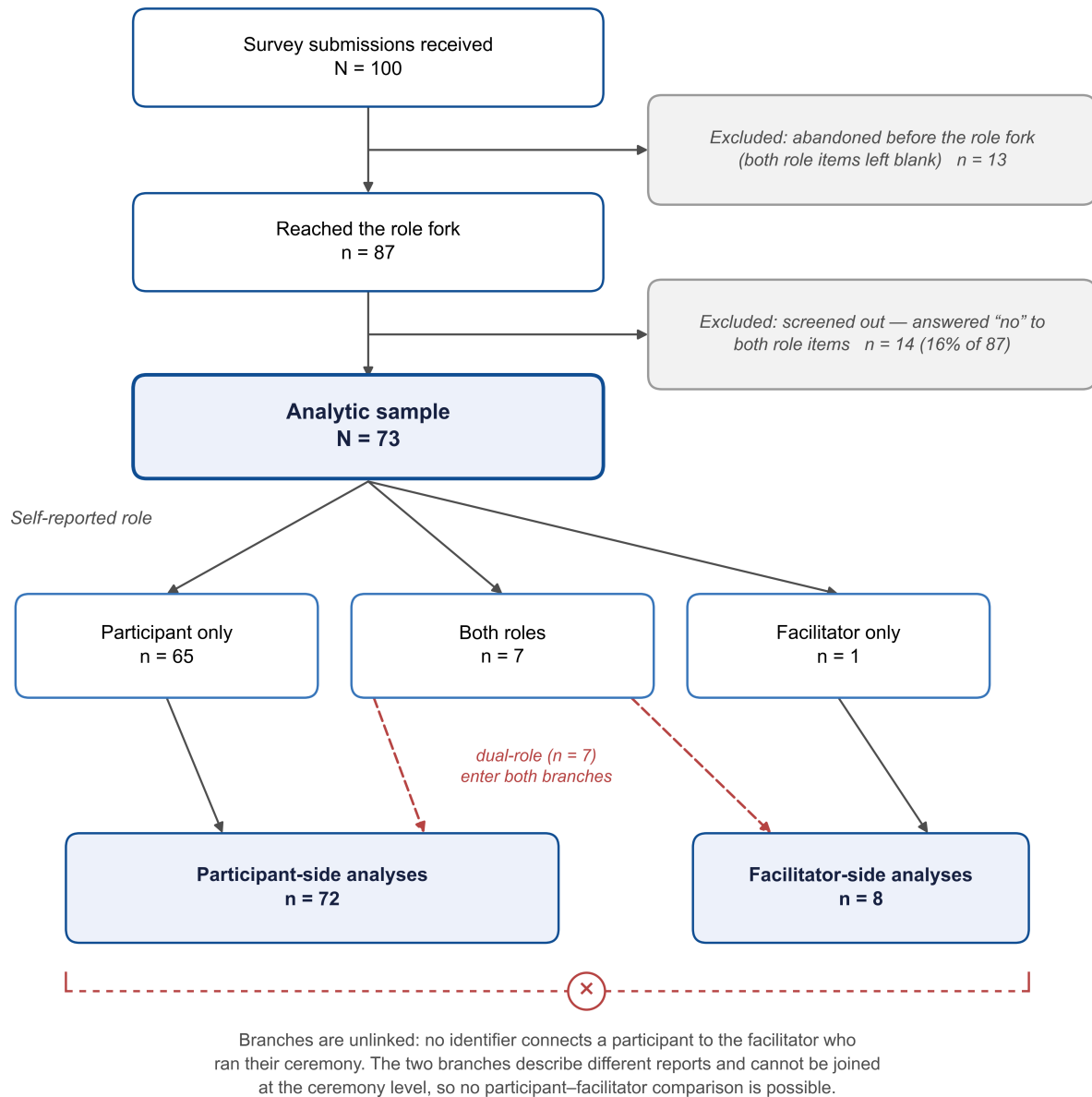
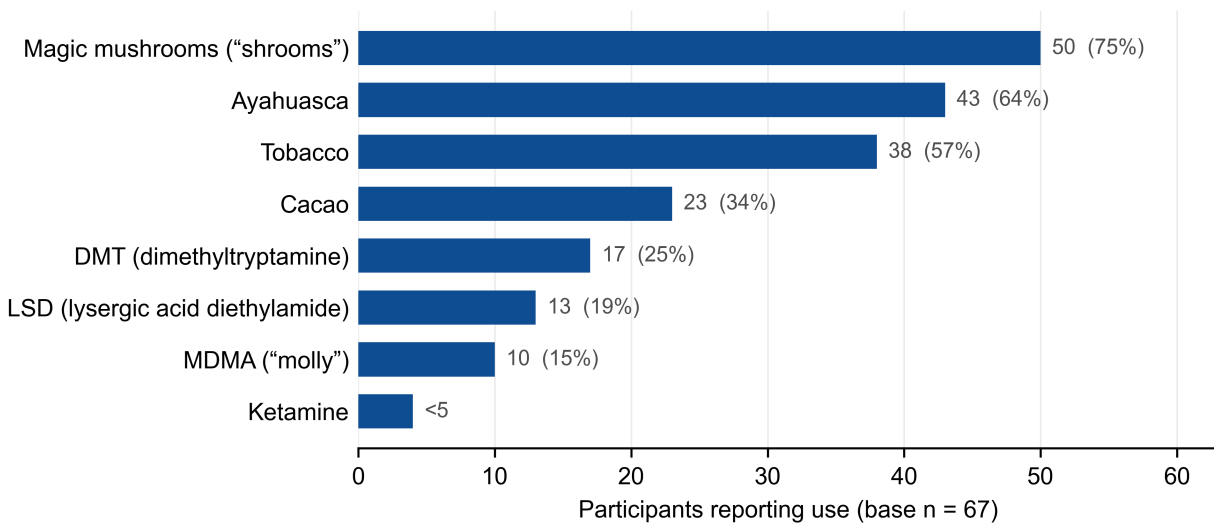


Figure 1: Sample construction and role structure. Of 100 submissions, 13 were abandoned before the role fork and 14 of the remaining 87 screened out by answering “no” to both role items, leaving an analytic sample of  $N = 73$ . Roles are not mutually exclusive: the 7 dual-role respondents enter both branches, giving 72 participant-side and 8 facilitator-side reports. The two branches are unlinkable — no identifier connects a participant to the facilitator who ran their ceremony — so no participant–facilitator comparison is possible.



*Select-multiple item: percentages are of the base and do not sum to 100%. Cells with  $n < 5$  masked.*

Figure 2: Substances used across all ceremonies attended, participant-reported (select-multiple; base  $n = 67$  participants answering). Bars are labelled with the count and the percentage of that base; because respondents could select more than one substance, percentages need not sum to 100%. Cells with  $n < 5$  are masked. Distinct from the facilitator-led substance variable and from the most-recent-ceremony substance variable.

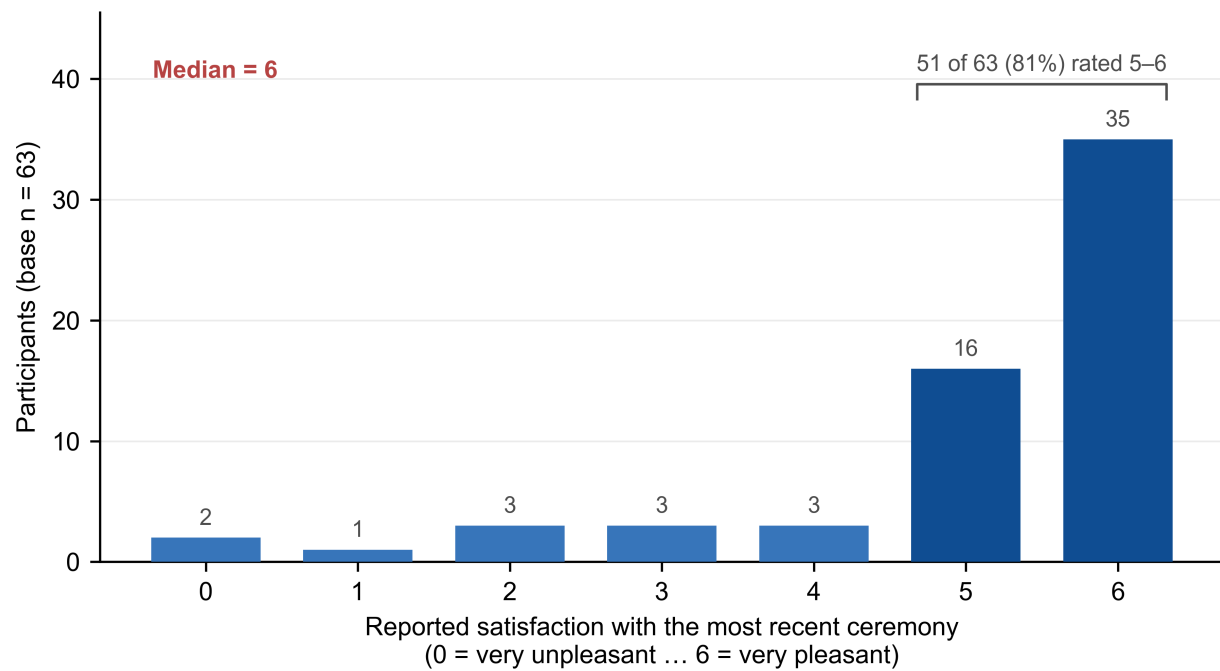
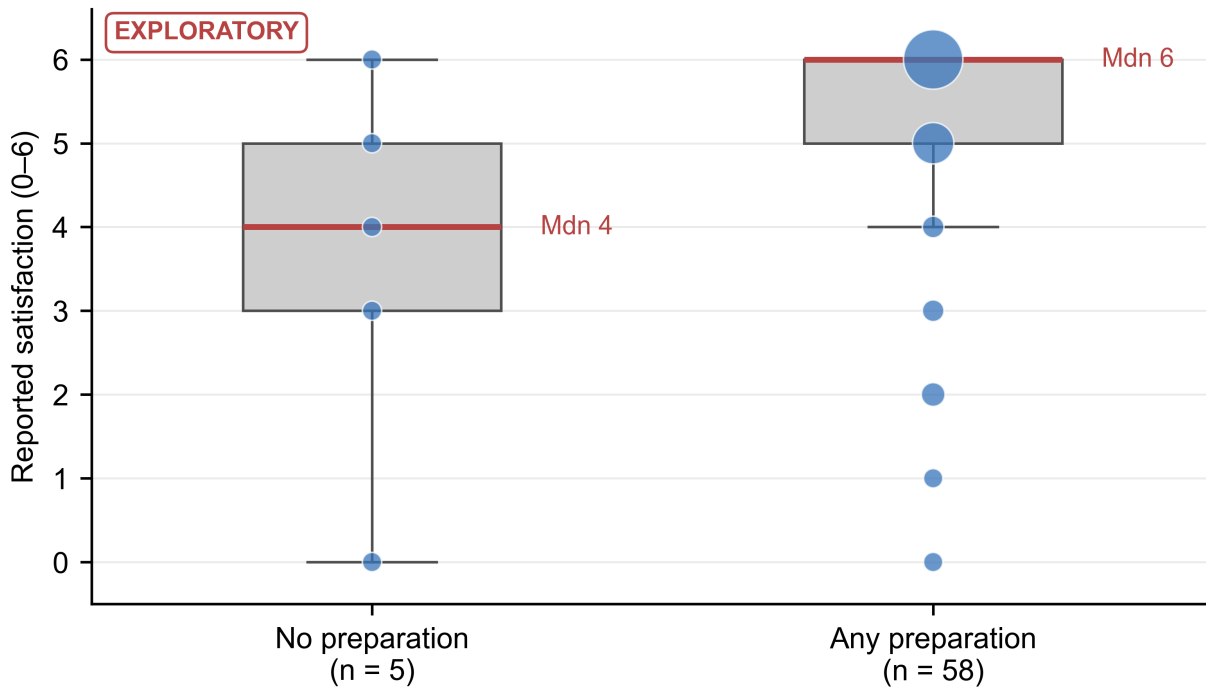


Figure 3: Distribution of reported satisfaction with the most recent ceremony (0 = very unpleasant to 6 = very pleasant). Base: 63 of 72 participants answering; the remainder is item nonresponse. Median 6, with 51 of 63 (81%) answering 5 or 6 — the ceiling that limits how much variance any predictor can explain.



Marker area is proportional to the number of respondents giving that score.  
The no-preparation group is  $n = 5$ : suggestive, not confirmatory.

Figure 4: Reported satisfaction by self-directed preparation status. Any preparation,  $n = 58$  answering the outcome (median 6, IQR 5–6); no preparation,  $n = 5$  (median 4, IQR 3–5). Marker area is proportional to the number of respondents giving each score, since scores tie heavily at the ceiling. **Exploratory:** the no-preparation group contains five respondents and this comparison is suggestive, not confirmatory.